Multiplying Fractions

Name _____ Date _____ No calculator Version 1

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KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · product *(otinga whakarea)*

A A fraction of an amount ●○○

Remember: Divide by the denominator, then multiply by the numerator: $\frac{3}{4}$ of 20 is $20 \div 4 \times 3 = 15$.

1. Work out.

a) $\frac{1}{2}$ of 18 = _____ **b)** $\frac{1}{4}$ of 20 = _____ **c)** $\frac{1}{3}$ of 27 = _____ **d)** $\frac{3}{4}$ of 20 = _____

e) $\frac{2}{3}$ of 15 = _____ **f)** $\frac{2}{5}$ of 30 = _____ **g)** $\frac{5}{6}$ of 24 = _____ **h)** $\frac{3}{8}$ of 40 = _____

i) $\frac{7}{10}$ of 60 = _____

B A whole number times a fraction ●●○

Watch out: Multiply the numerator only: $3 \times \frac{2}{5} = \frac{6}{5} = 1\frac{1}{5}$. The denominator does not change.

2. Work out. Give each answer in its simplest form, as a mixed number if it is more than 1.

a) $\frac{1}{5} \times 3$ **b)** $\frac{2}{7} \times 3$ **c)** $\frac{3}{4} \times 5$ **d)** $\frac{2}{3} \times 4$

e) $\frac{5}{6} \times 3$ **f)** $\frac{3}{8} \times 4$ **g)** $\frac{7}{10} \times 5$ **h)** $\frac{4}{9} \times 6$

3. Find each fraction of an amount. The answers are not whole numbers.

a) $\frac{1}{2}$ of 7 **b)** $\frac{2}{3}$ of 10 **c)** $\frac{3}{4}$ of 9 **d)** $\frac{2}{5}$ of 12

C A fraction times a fraction ●●○

Watch out: You do not need a common denominator to multiply. Multiply the numerators, and multiply the denominators.

4. Work out. Give each answer in its simplest form.

a) $\frac{1}{2} \times \frac{1}{3}$ **b)** $\frac{1}{4} \times \frac{1}{5}$ **c)** $\frac{2}{3} \times \frac{1}{5}$ **d)** $\frac{3}{4} \times \frac{1}{2}$

e) $\frac{2}{3} \times \frac{3}{4}$ **f)** $\frac{3}{5} \times \frac{2}{7}$ **g)** $\frac{4}{5} \times \frac{5}{8}$ **h)** $\frac{5}{6} \times \frac{3}{10}$

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D Multiplying by a mixed number ●●●

> **Remember:** Write the mixed number as an improper fraction first: $4 \times 1\frac{1}{2} = 4 \times \frac{3}{2} = \frac{12}{2} = 6$.

5. Work out. Give each answer in its simplest form.

a) $4 \times 1\frac{1}{2}$ **b)** $3 \times 2\frac{1}{3}$ **c)** $2 \times 1\frac{3}{4}$ **d)** $5 \times 1\frac{1}{5}$

e) $3 \times 1\frac{2}{5}$ **f)** $6 \times 2\frac{1}{4}$ **g)** $2 \times 3\frac{2}{3}$ **h)** $4 \times 1\frac{5}{6}$

E Finding the whole ●●●

> **Remember:** Work back to one part first. If $\frac{3}{4}$ of a number is 12, then $\frac{1}{4}$ of it is $12 \div 3 = 4$, so the number is $4 \times 4 = 16$.

6. Find the number.

a) $\frac{1}{3}$ of a number is 7 **b)** $\frac{1}{5}$ of a number is 9 **c)** $\frac{3}{4}$ of a number is 12

d) $\frac{2}{3}$ of a number is 10 **e)** $\frac{2}{5}$ of a number is 8 **f)** $\frac{5}{6}$ of a number is 20

F Problems ●●●

7. A class has 28 students. $\frac{3}{4}$ of them have a pet. How many students have a pet?

8. A recipe needs $2\frac{1}{2}$ cups of flour. How many cups of flour are needed to make the recipe 6 times?

9. Mele has read $\frac{2}{5}$ of her book. She has read 14 pages. How many pages does the book have?

★ Extension

10. Work out $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4}$.

11. Tom has $\frac{2}{3}$ of a cake. He eats $\frac{1}{2}$ of what he has. What fraction of the whole cake does he eat?

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Multiplying Fractions

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · product *(otinga whakarea)*

A A fraction of an amount ●○○

Remember: Divide by the denominator, then multiply by the numerator: $\frac{3}{4}$ of 20 is $20 \div 4 \times 3 = 15$.

1. Work out.

a) $\frac{1}{2}$ of 26 = _____ **b)** $\frac{1}{4}$ of 28 = _____ **c)** $\frac{1}{3}$ of 21 = _____ **d)** $\frac{3}{4}$ of 16 = _____

e) $\frac{2}{3}$ of 18 = _____ **f)** $\frac{3}{5}$ of 20 = _____ **g)** $\frac{5}{6}$ of 30 = _____ **h)** $\frac{3}{8}$ of 32 = _____

i) $\frac{9}{10}$ of 50 = _____

B A whole number times a fraction ●●○

Watch out: Multiply the numerator only: $3 \times \frac{2}{5} = \frac{6}{5} = 1\frac{1}{5}$. The denominator does not change.

2. Work out. Give each answer in its simplest form, as a mixed number if it is more than 1.

a) $\frac{1}{7} \times 4$ **b)** $\frac{2}{9} \times 4$ **c)** $\frac{3}{4} \times 7$ **d)** $\frac{2}{3} \times 5$

e) $\frac{5}{6} \times 4$ **f)** $\frac{5}{8} \times 4$ **g)** $\frac{3}{10} \times 5$ **h)** $\frac{2}{9} \times 6$

3. Find each fraction of an amount. The answers are not whole numbers.

a) $\frac{1}{2}$ of 9 **b)** $\frac{2}{3}$ of 8 **c)** $\frac{3}{4}$ of 11 **d)** $\frac{3}{5}$ of 12

C A fraction times a fraction ●●○

Watch out: You do not need a common denominator to multiply. Multiply the numerators, and multiply the denominators.

4. Work out. Give each answer in its simplest form.

a) $\frac{1}{2} \times \frac{1}{5}$ **b)** $\frac{1}{3} \times \frac{1}{4}$ **c)** $\frac{3}{4} \times \frac{1}{5}$ **d)** $\frac{2}{3} \times \frac{1}{2}$

e) $\frac{3}{4} \times \frac{2}{5}$ **f)** $\frac{2}{5} \times \frac{3}{7}$ **g)** $\frac{3}{4} \times \frac{4}{9}$ **h)** $\frac{5}{6} \times \frac{2}{5}$

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D Multiplying by a mixed number ●●●

> **Remember:** Write the mixed number as an improper fraction first: $4 \times 1\frac{1}{2} = 4 \times \frac{3}{2} = \frac{12}{2} = 6$.

5. Work out. Give each answer in its simplest form.

a) $6 \times 1\frac{1}{2}$ **b)** $3 \times 1\frac{2}{3}$ **c)** $2 \times 2\frac{1}{4}$ **d)** $5 \times 1\frac{2}{5}$

e) $3 \times 1\frac{3}{4}$ **f)** $4 \times 2\frac{1}{8}$ **g)** $2 \times 2\frac{2}{3}$ **h)** $4 \times 1\frac{1}{6}$

E Finding the whole ●●●

> **Remember:** Work back to one part first. If $\frac{3}{4}$ of a number is 12, then $\frac{1}{4}$ of it is $12 \div 3 = 4$, so the number is $4 \times 4 = 16$.

6. Find the number.

a) $\frac{1}{4}$ of a number is 6 **b)** $\frac{1}{6}$ of a number is 8 **c)** $\frac{3}{4}$ of a number is 15

d) $\frac{2}{3}$ of a number is 14 **e)** $\frac{2}{5}$ of a number is 12 **f)** $\frac{5}{6}$ of a number is 25

F Problems ●●●

7. A class has 27 students. $\frac{2}{3}$ of them have a pet. How many students have a pet?

8. A recipe needs $3\frac{1}{2}$ cups of flour. How many cups of flour are needed to make the recipe 4 times?

9. Mele has read $\frac{3}{5}$ of her book. She has read 24 pages. How many pages does the book have?

★ Extension

10. Work out $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{5}$.

11. Tom has $\frac{3}{4}$ of a cake. He eats $\frac{1}{3}$ of what he has. What fraction of the whole cake does he eat?

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